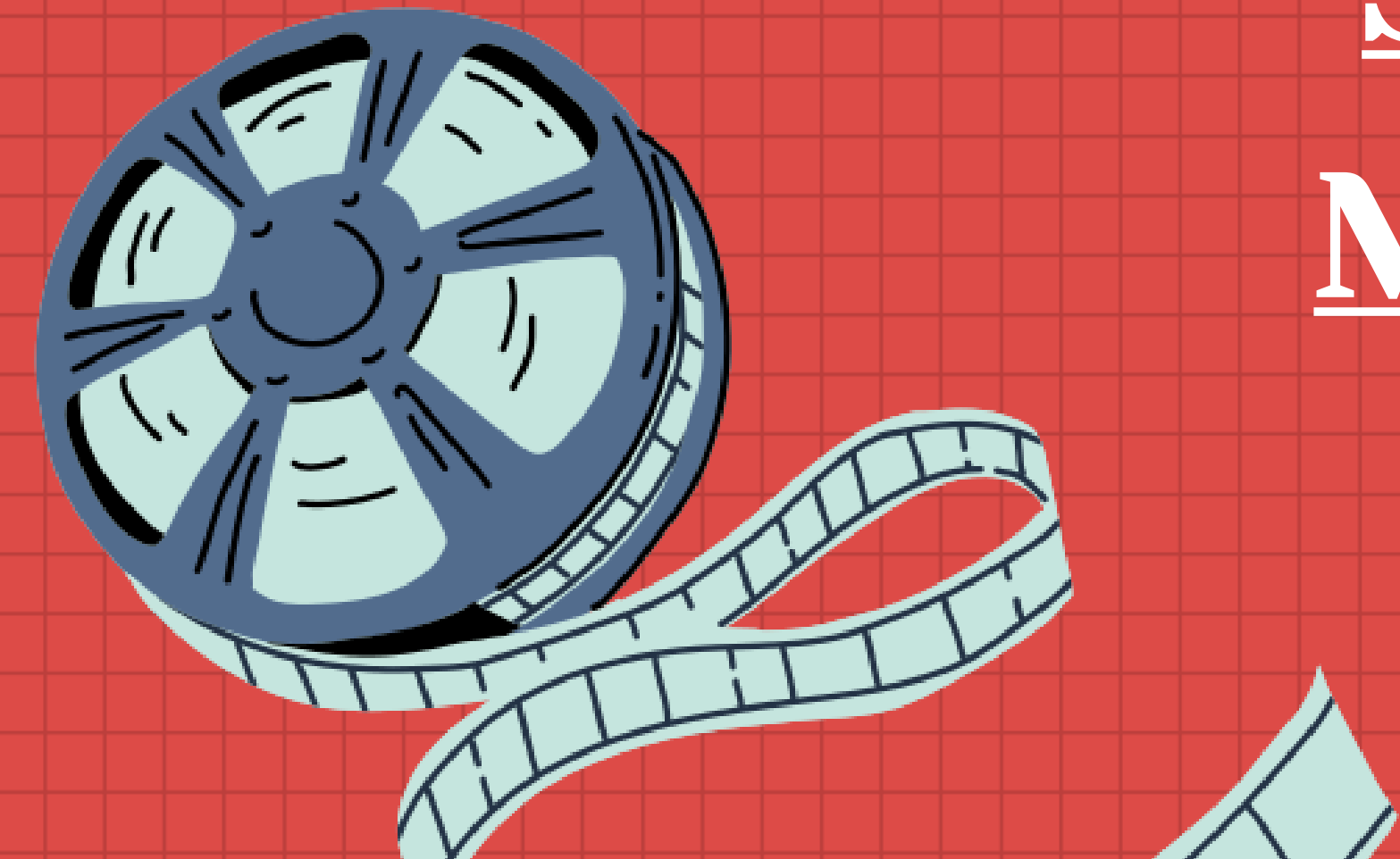
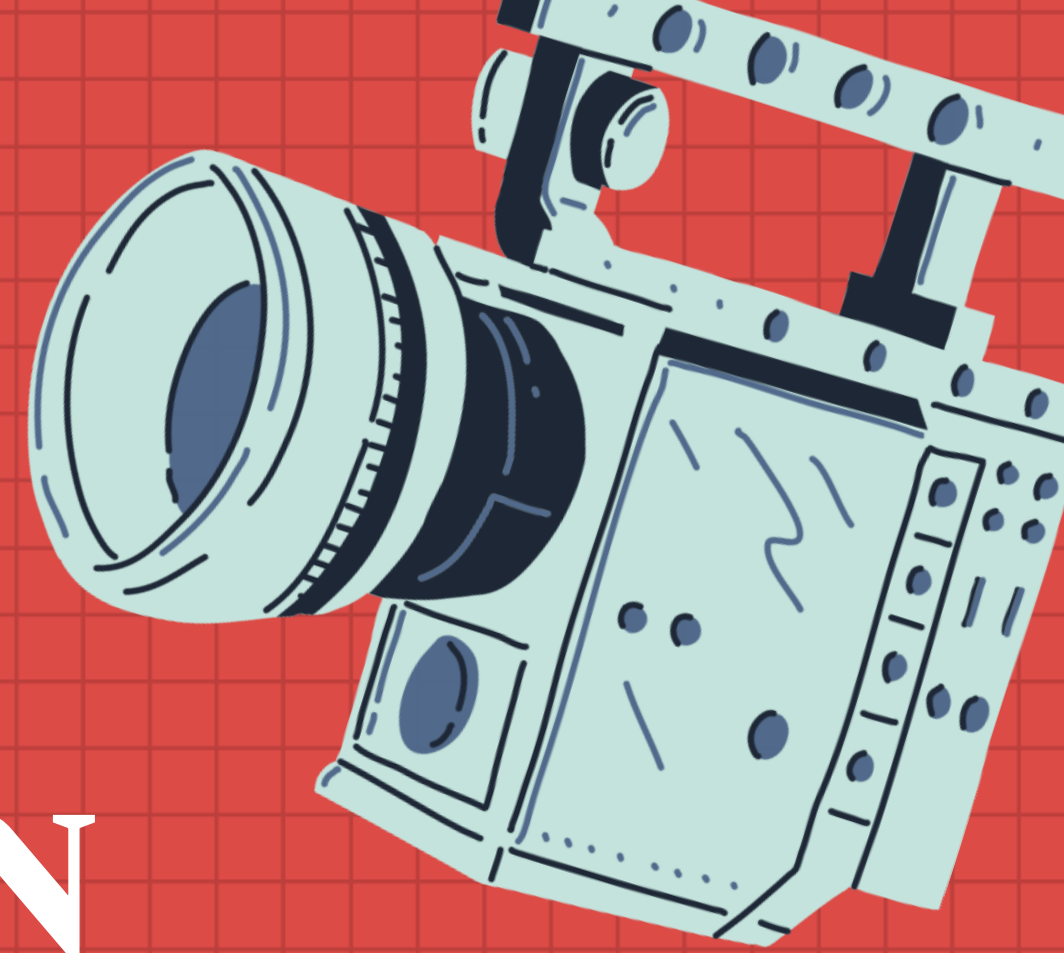


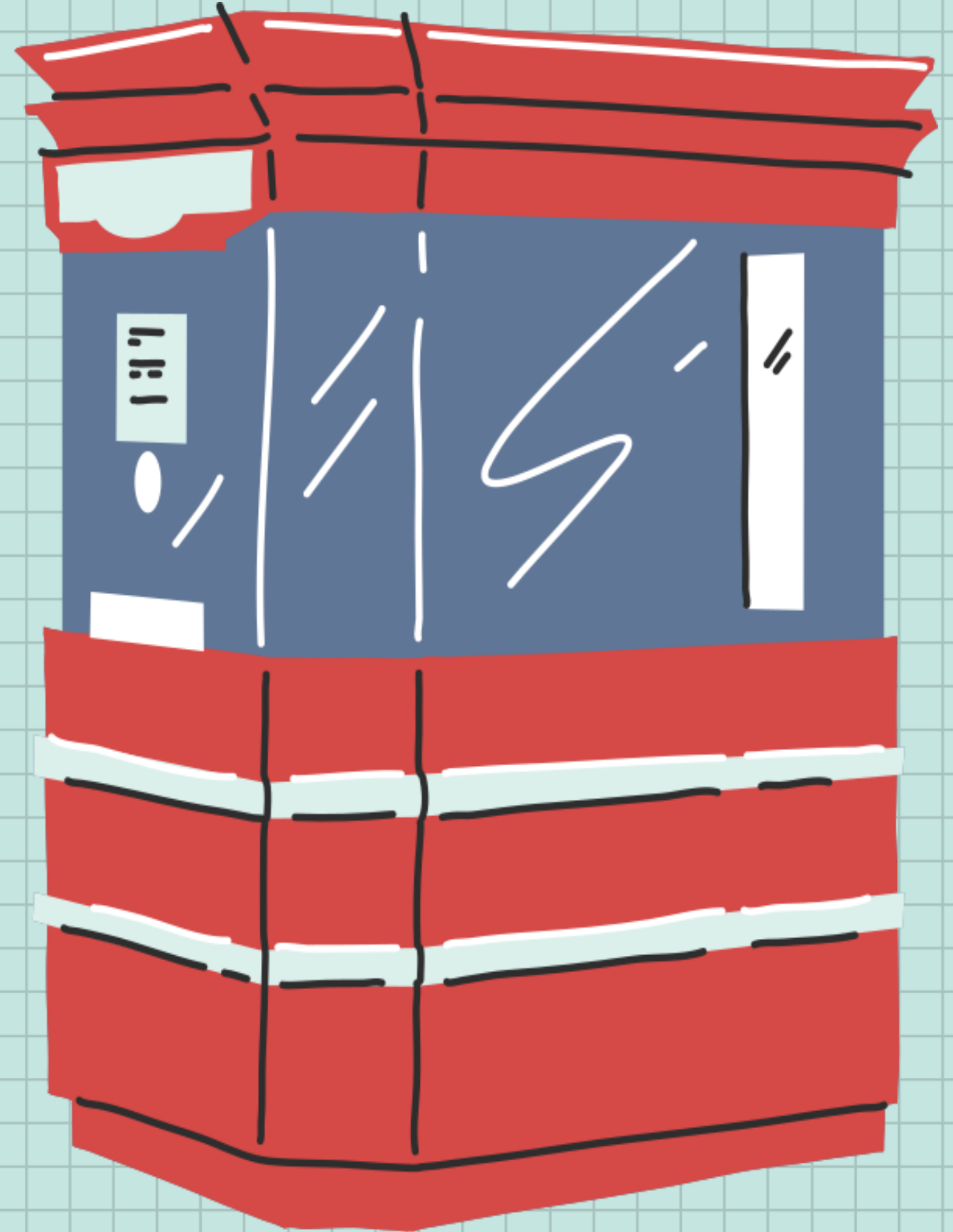


MOVIE
RECCOMENDATION
SYSTEM:
MOVIE101



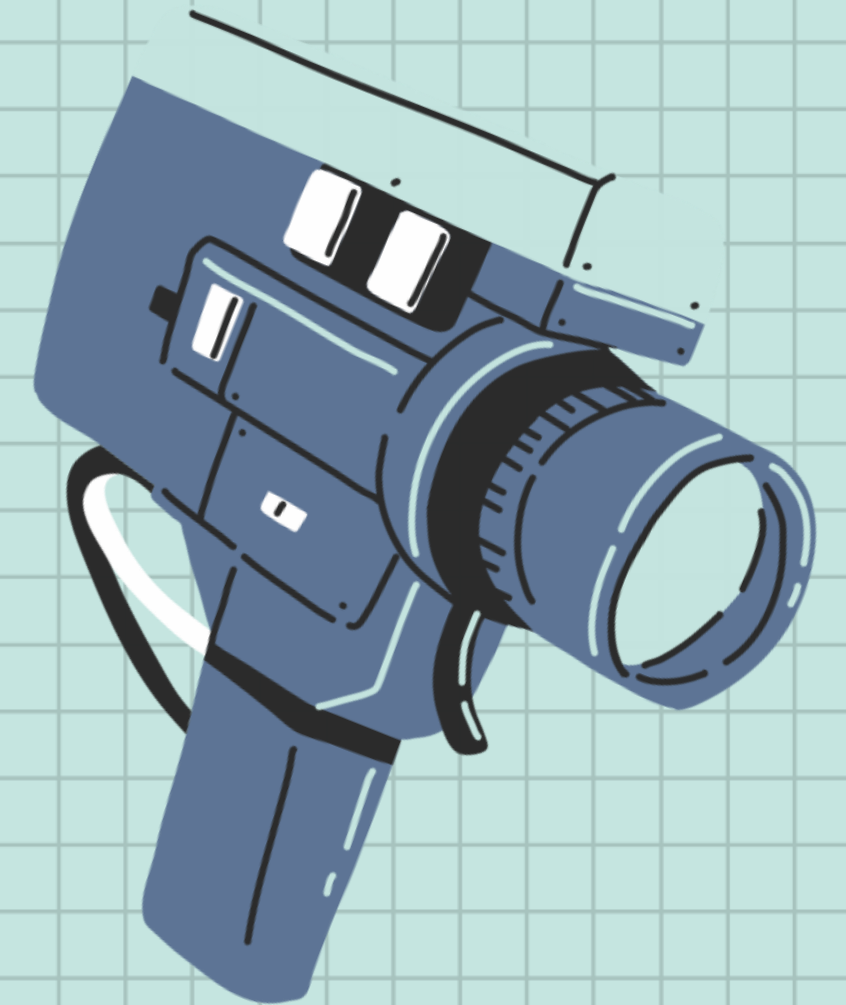
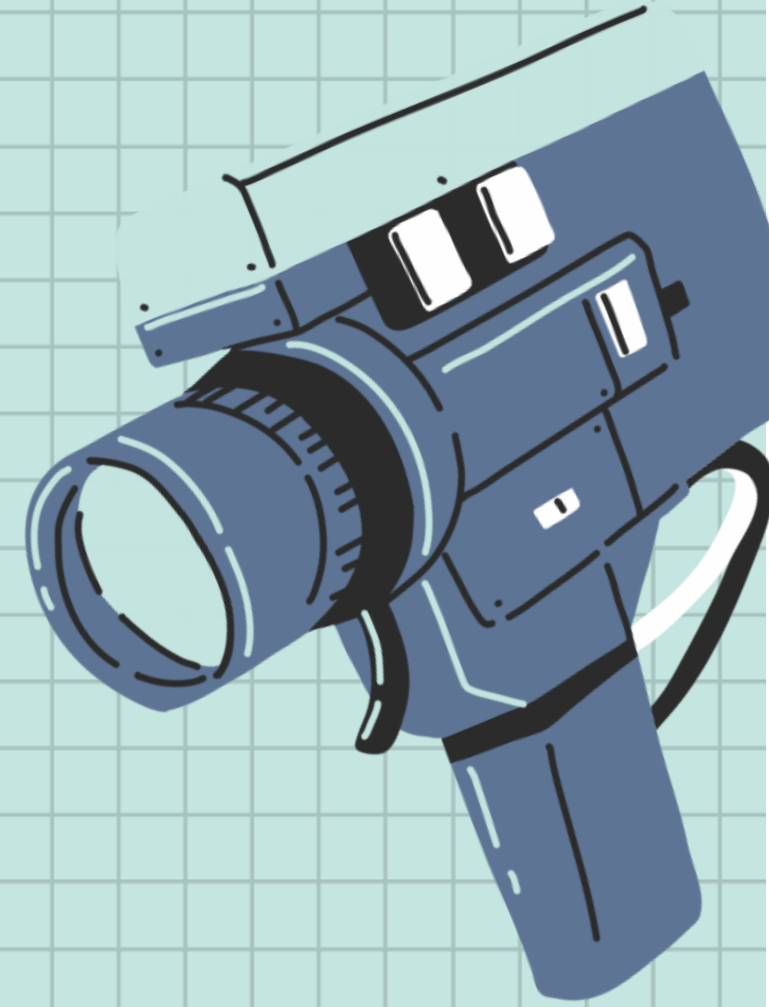
GROUP MEMBERS

- Sylvia Manono
- Amos Kipngetich
- Charles Ndegwa
- Sandra Koech
- Angela Maina
- Gloria Tisnanga
- Alex Miningwa



CONTENTS

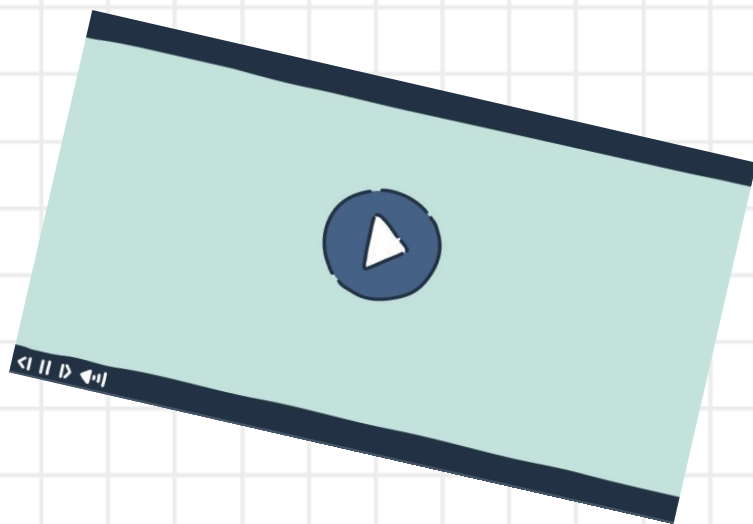
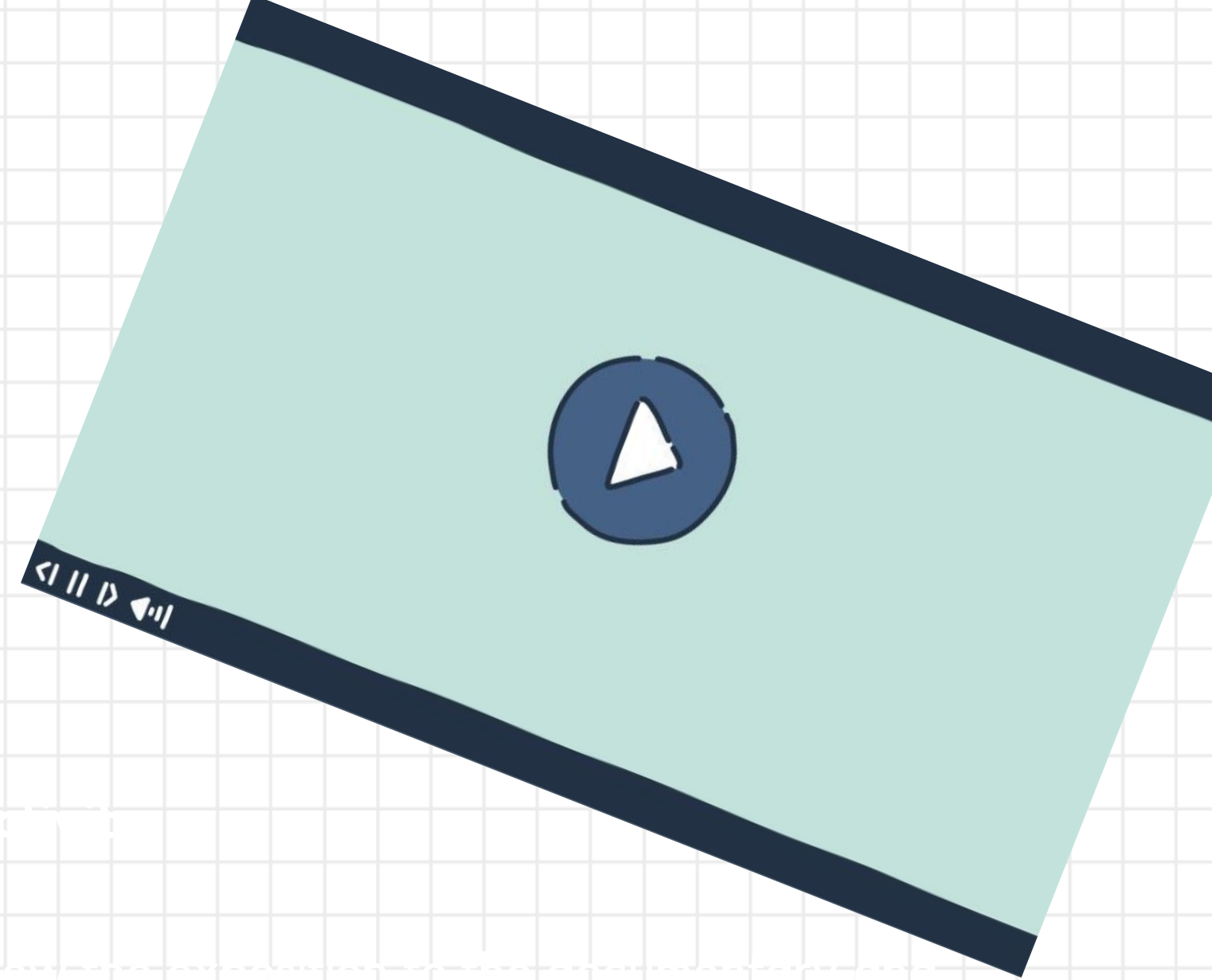
- Overview
- Problem statement, Objectives, Challenges
- Data Understanding
- EDA
- Modelling
- Evaluation
- Recommendations



1. OVERVIEW

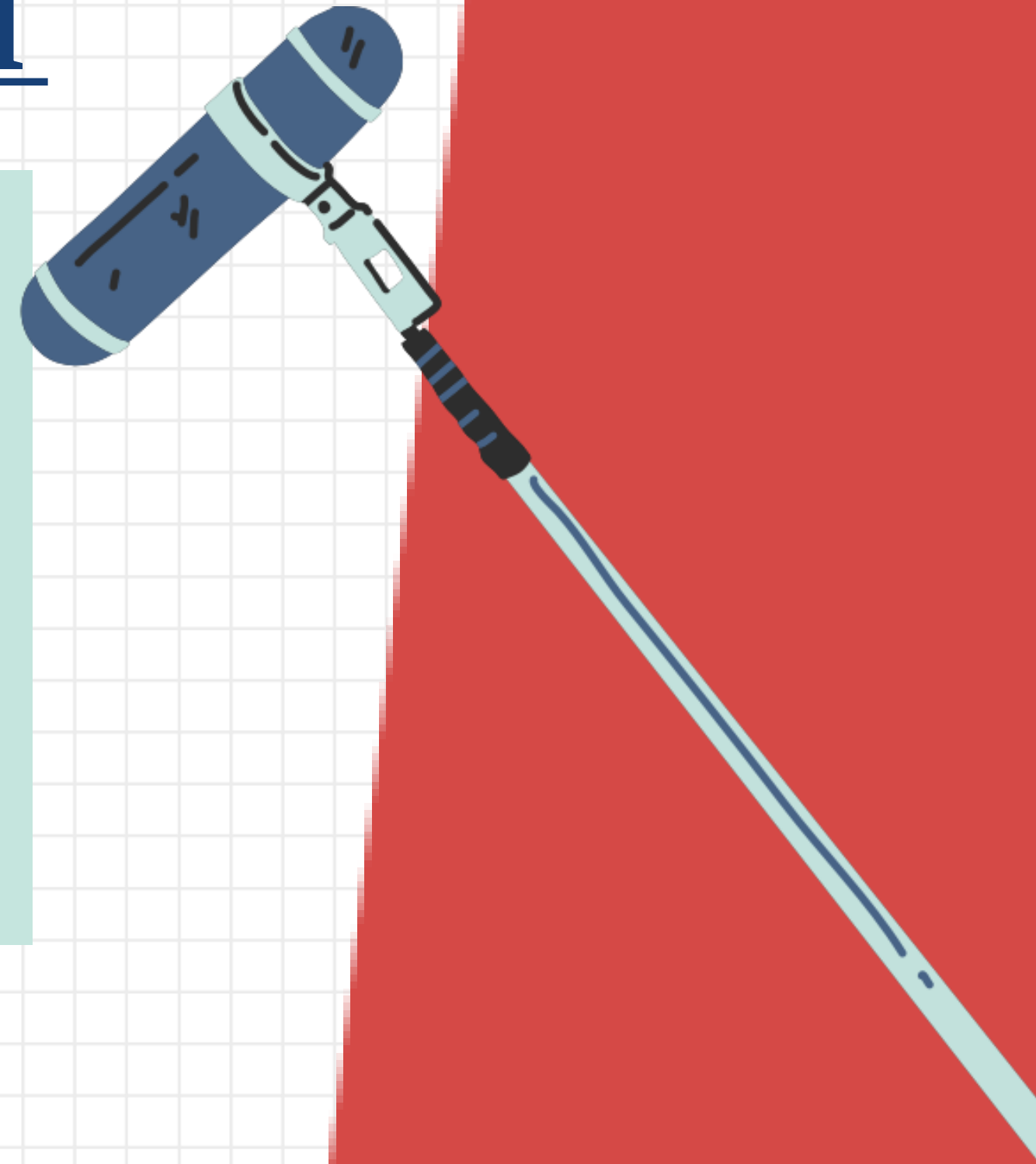
Movie101 is a rapidly growing movie streaming platform that aims to improve the user satisfaction by enhancing engagement on the platform. This could be achieved by offering personalized recommendations and encouraging users to discover more movies thus extending the viewing time.

Collaborative filtering, content-based filtering and hybrid methods will be used to create the recommendation system.



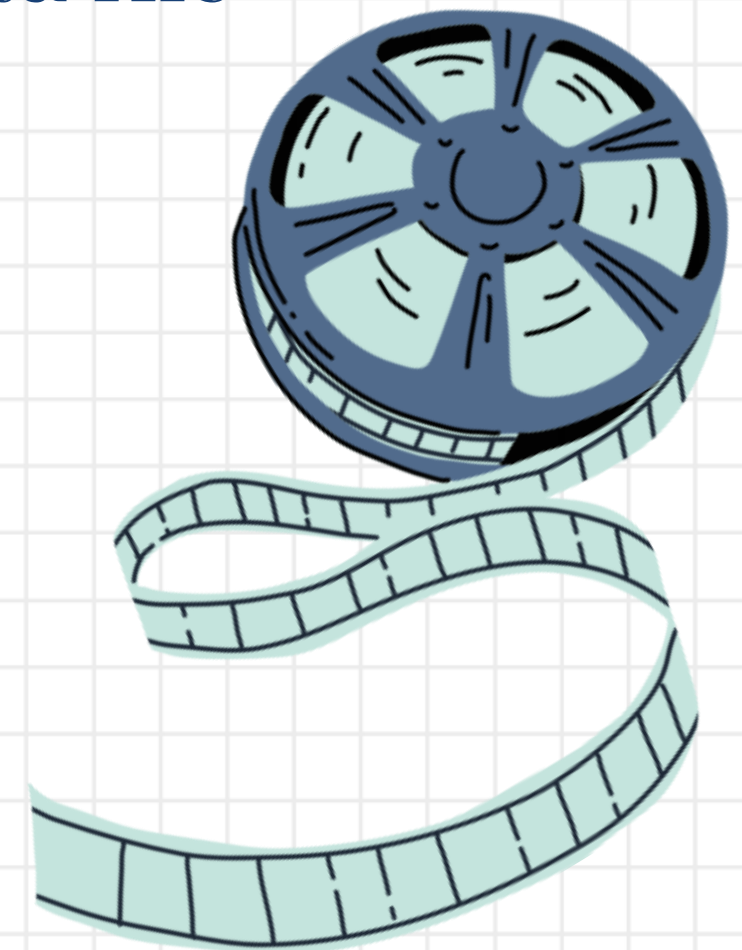
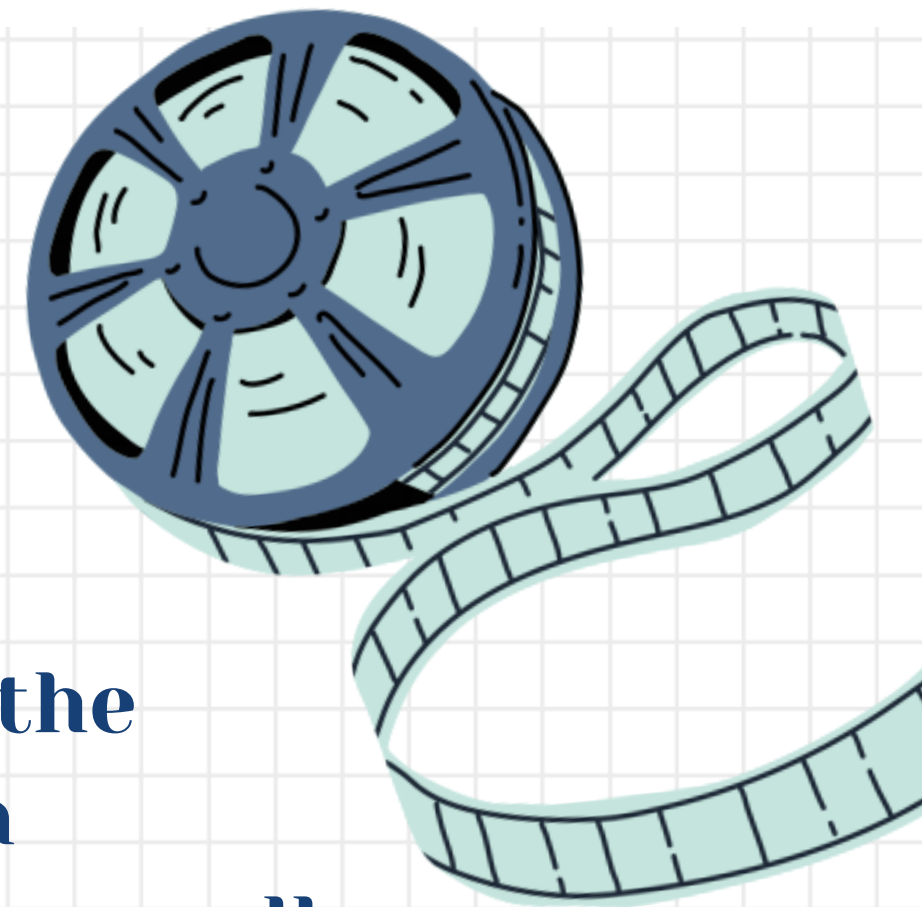
2. PROBLEM STATEMENT

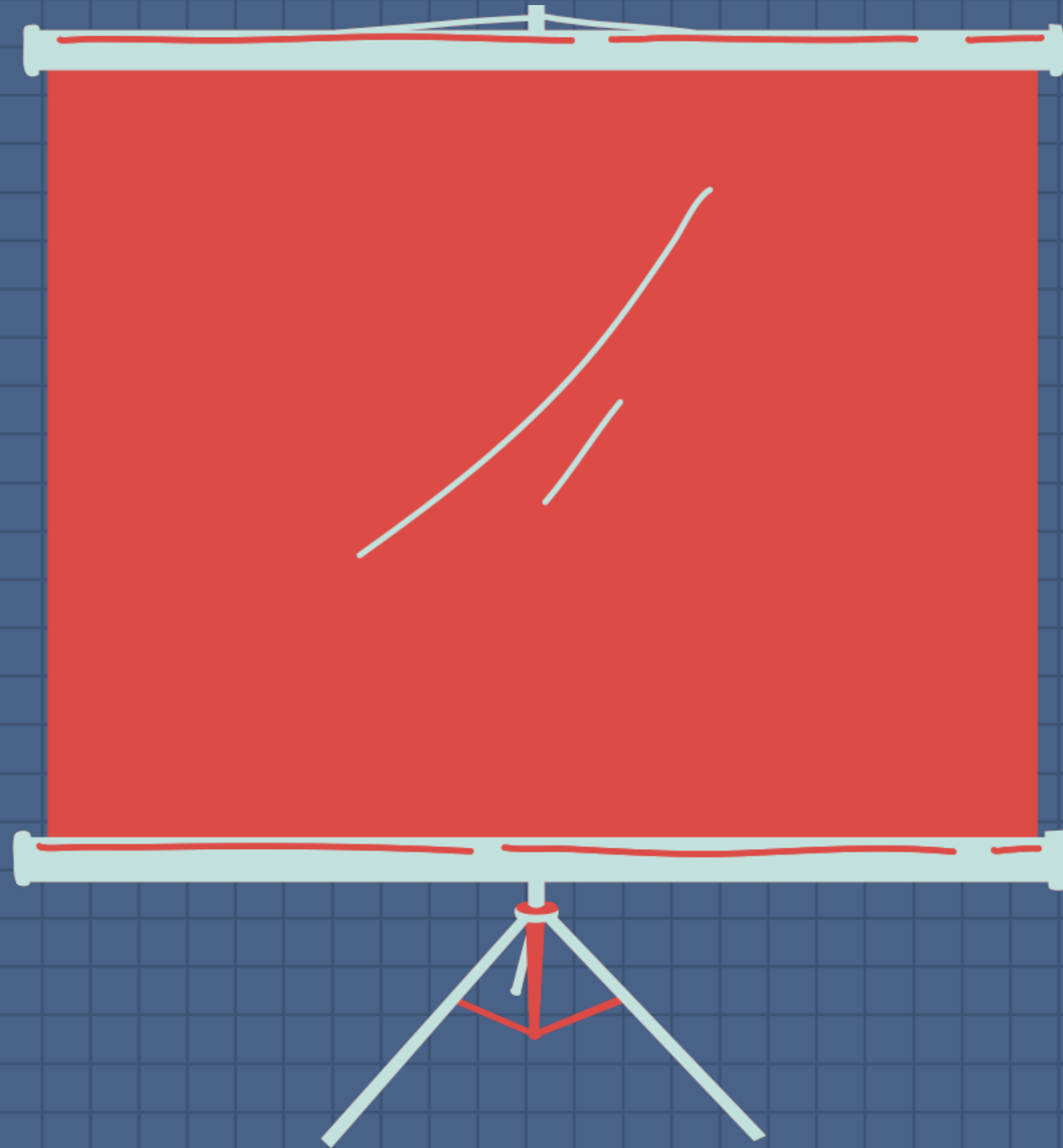
- The current platform's system fails to deliver adequate recommendations to existing users resulting in low levels of engagement, satisfaction and thus retention.**
- New users also face the disadvantage of not being able to get personalized recommendations based on their interests.**



3. DATA UNDERSTANDING

- The data used was sourced from MovieLens dataset <https://grouplens.org/datasets/movielens/latest/> from the GroupLens research lab at the University of Minnesota
- Given the constraints on computational resources, the small dataset of 100,000 user ratings was used.
- There were 4 csv files with different columns in the data file
 - a) movies.csv
 - b) Ratings.csv
 - c) Tags.csv
 - d) Links.csv



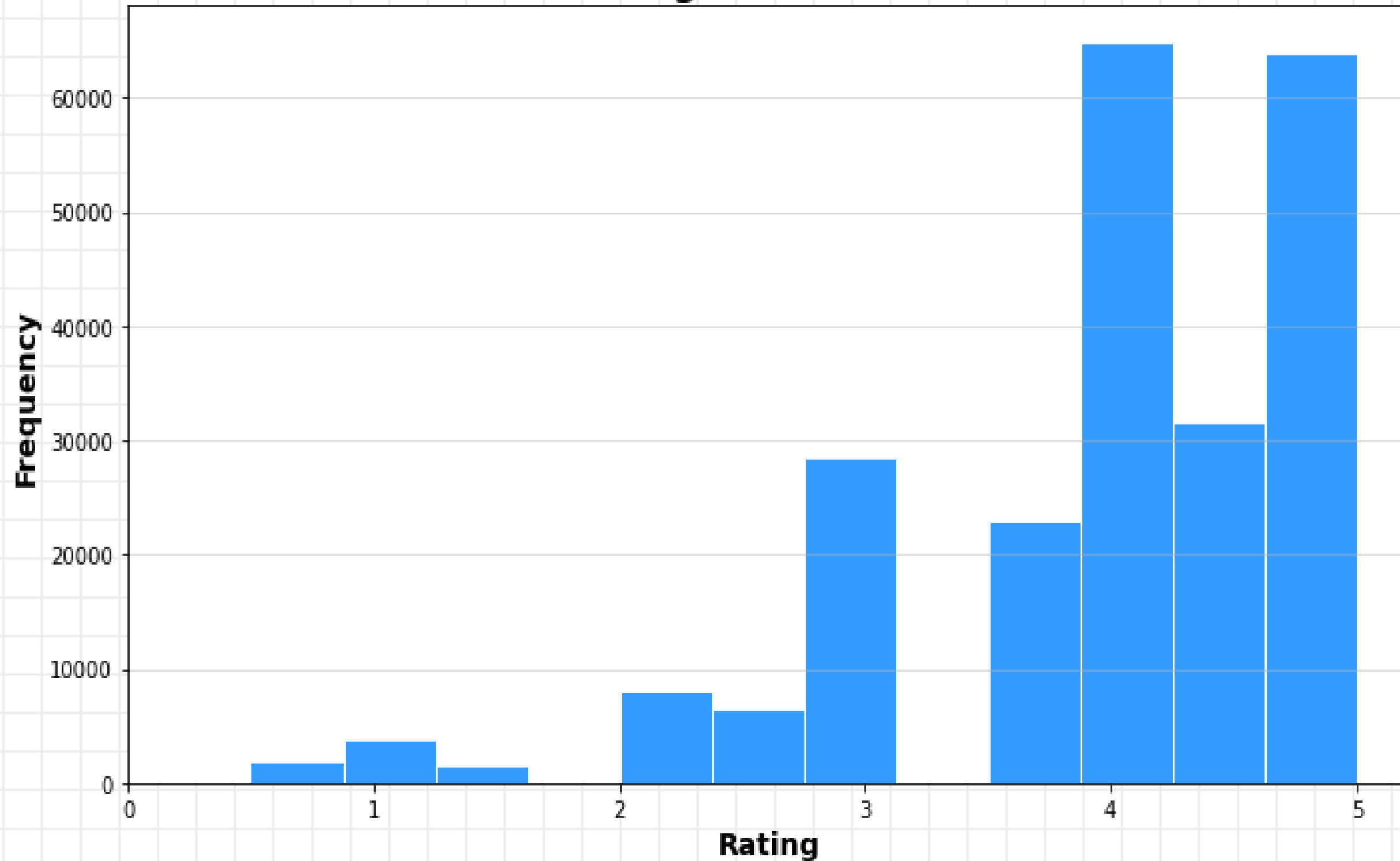


4. EXPLORATORY DATA ANALYSIS (EDA)



4.1 Distribution of ratings

Ratings Distribution

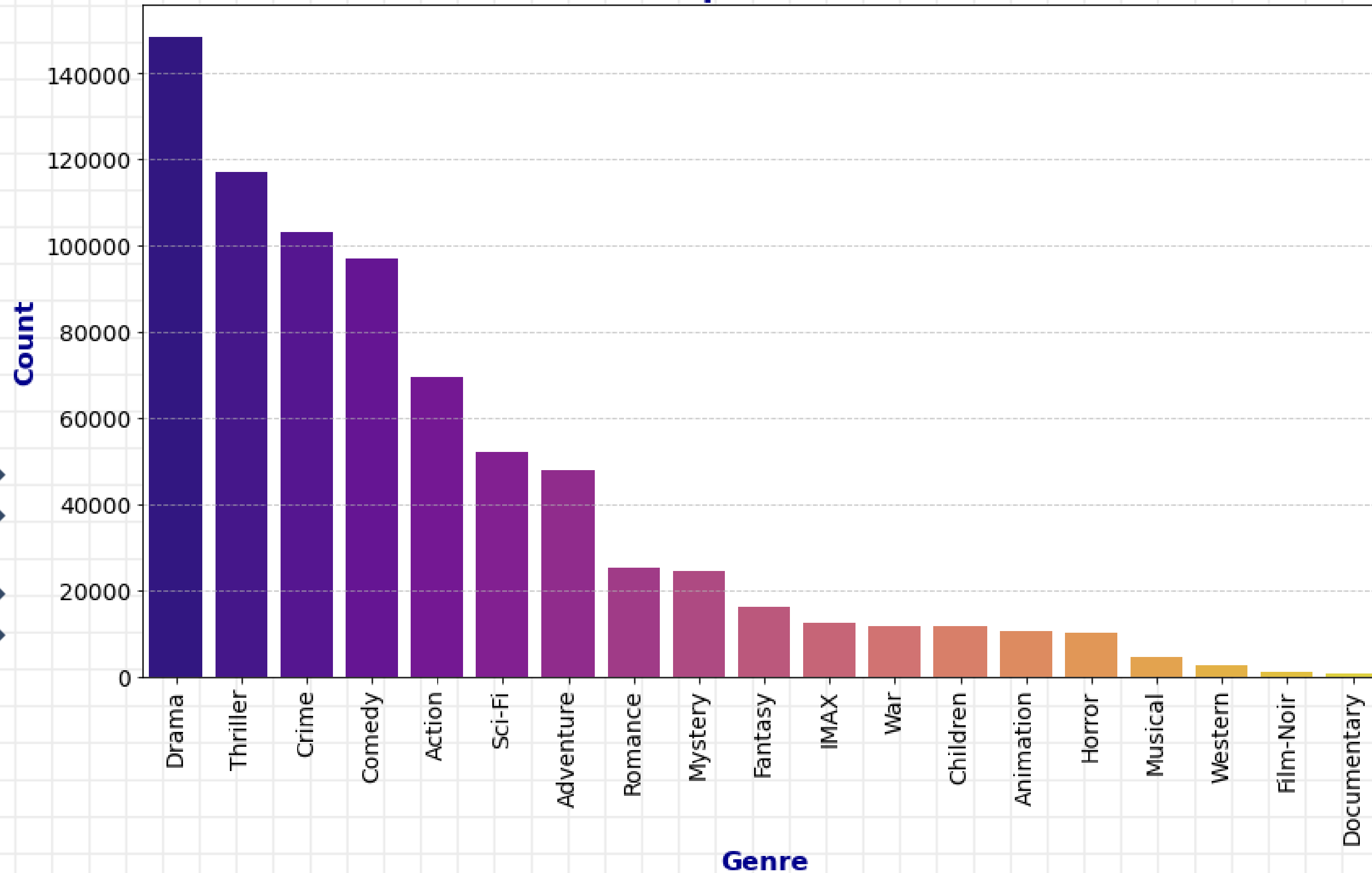


- The graph seems to be clustered more to the right
- This means that Higher ratings (4 & 5) are more common. Lower ratings are less frequent (1 & 2)

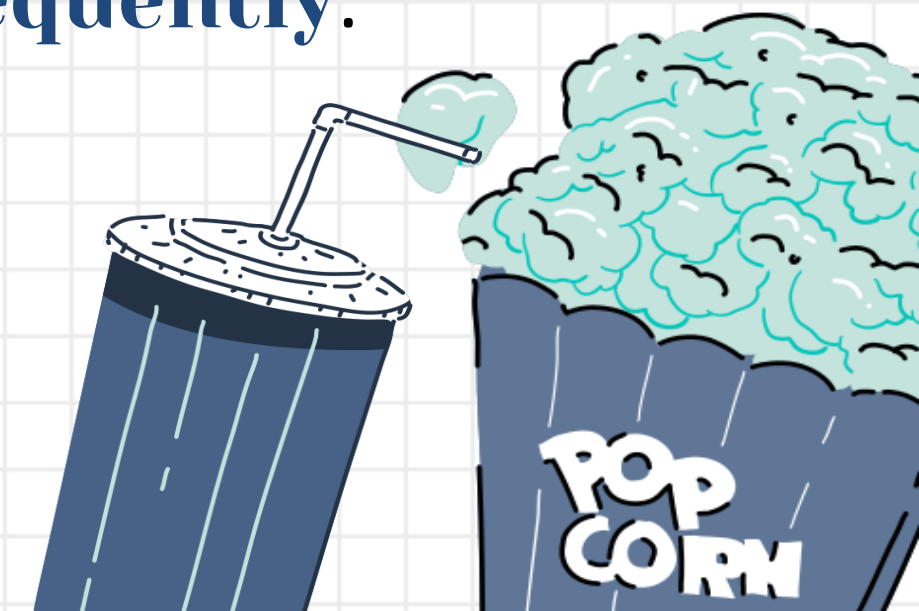


4.2 Top Genres

Top Genres

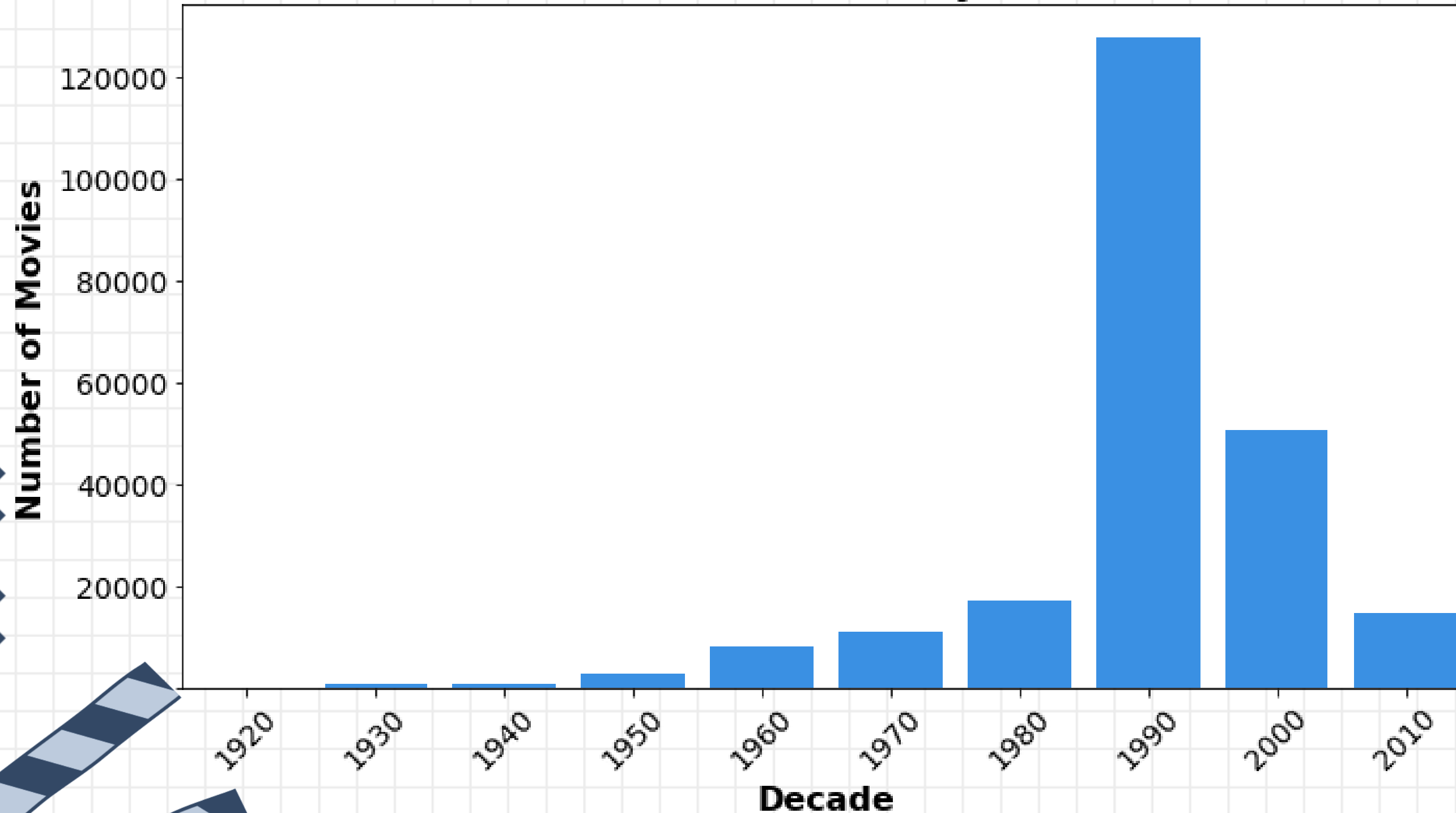


- The chart suggests that a few genres on the left are much more common than the others, showing a typical long-tail distribution, where a small number of genres are very popular, while many others occur far less frequently.

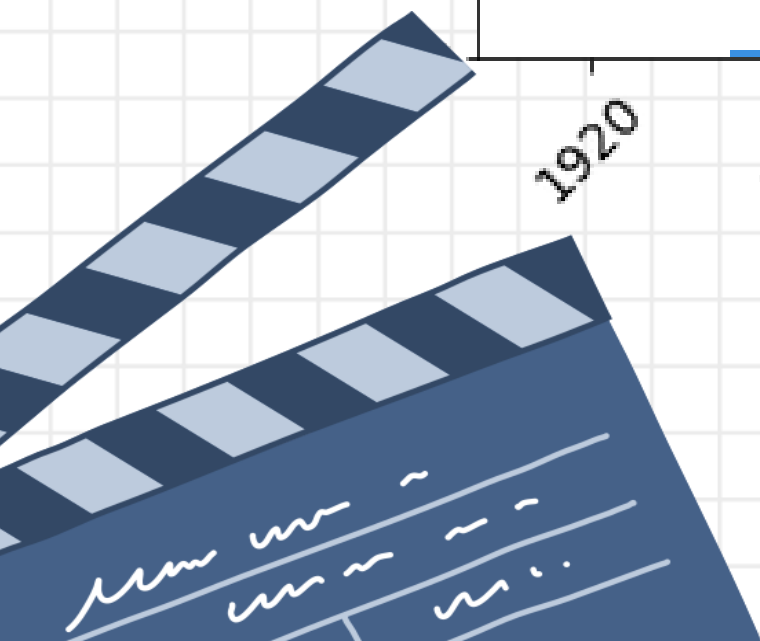


4.3 Count of movies by decade

Number of Movies by Decade

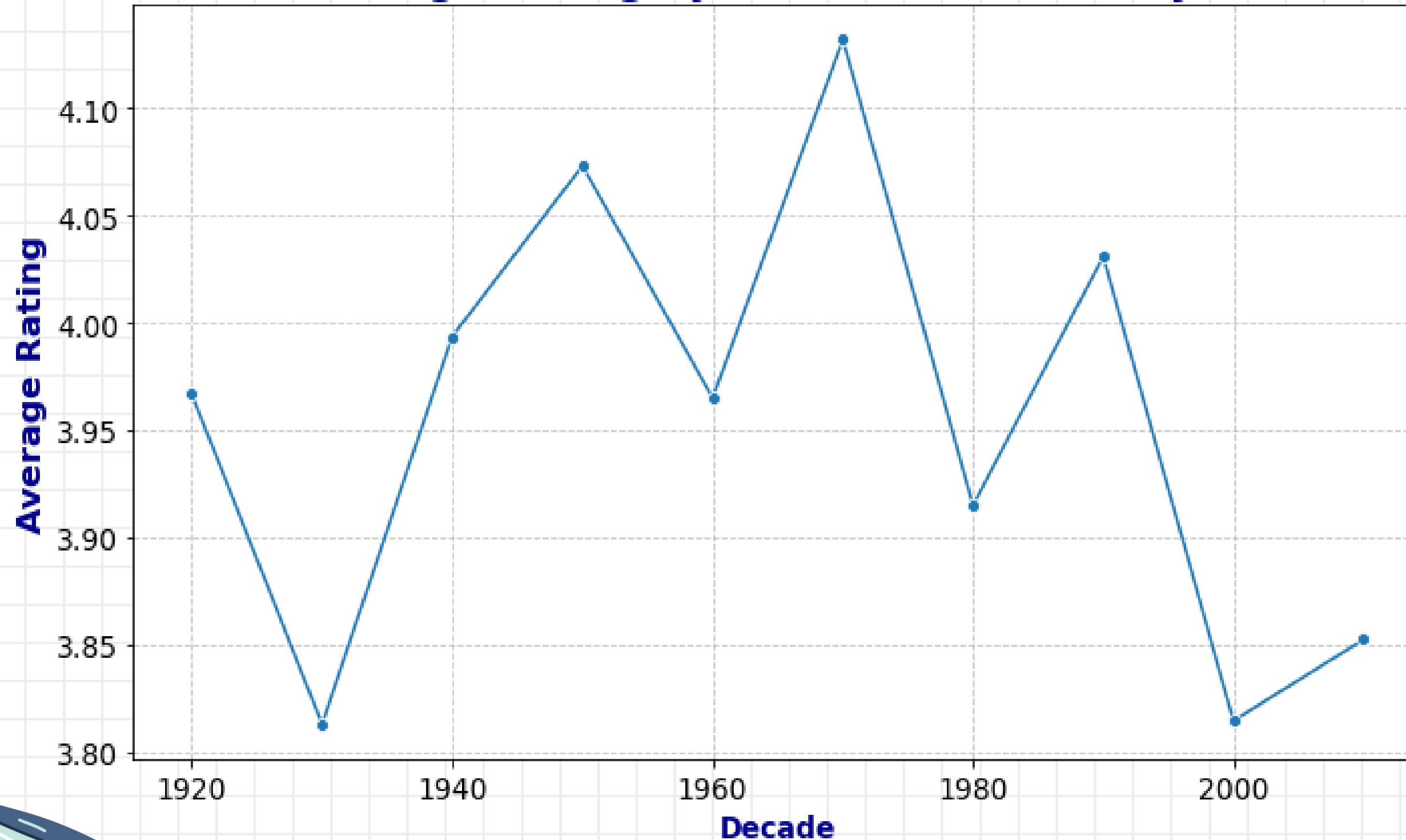


- The chart suggests that the 1930s had far fewer movies.
- 1990s saw a massive spike, with over 120,000 films produced
- Movie production dropped significantly between 2000 and 2010.



4.4 Average rating overtime

Average Rating by Decade- Trend Analysis



- The trend shows that the ratings exhibit a cyclical behavior with periods of rising and falling.
- The highest peak indicate the best-rated decade.
- The sharp decline towards the most recent decades may indicate changing audience preferences,

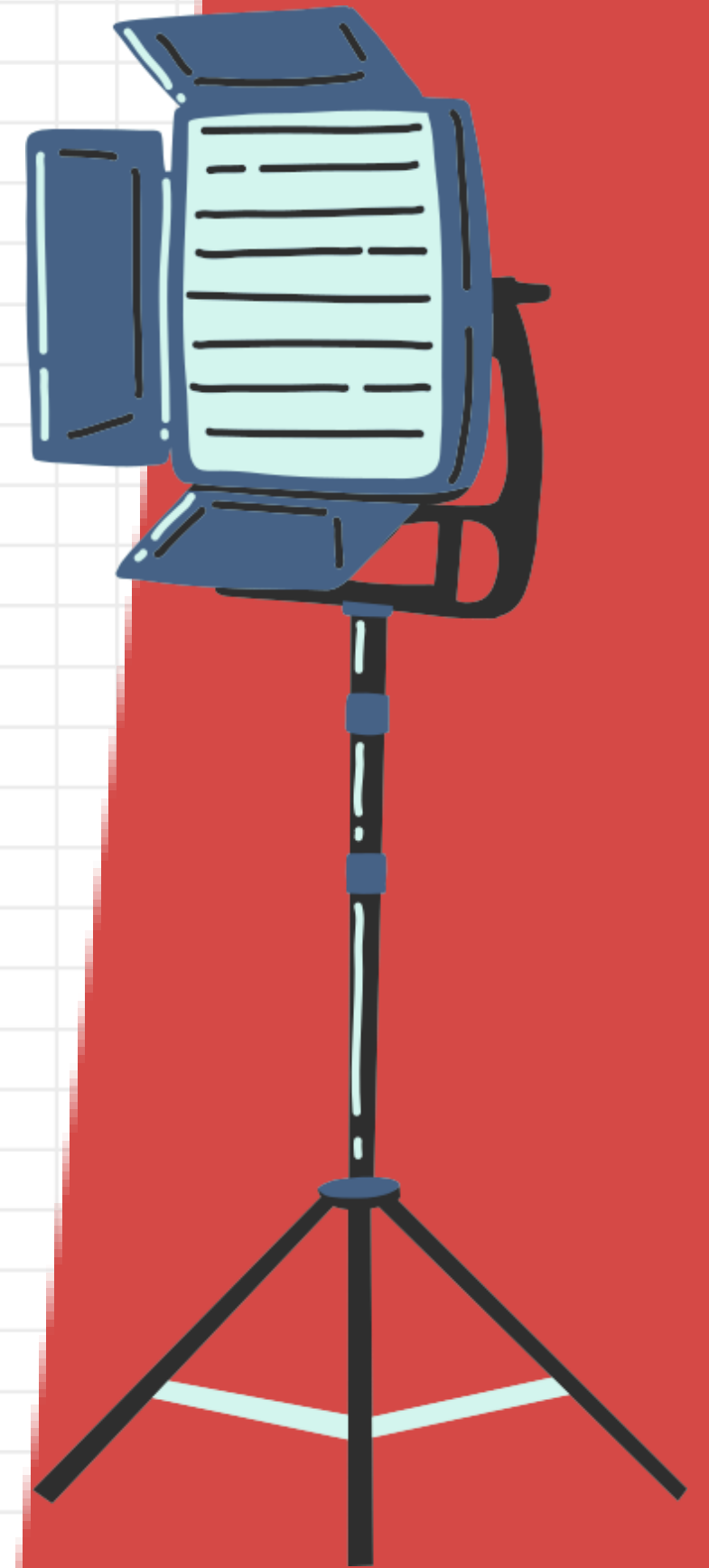
5. MODELLING

To conduct the evaluation, the dataset was split into training (80%) and test(20%) sets. This is to ensure that the model's performance is assessed on unseen data.

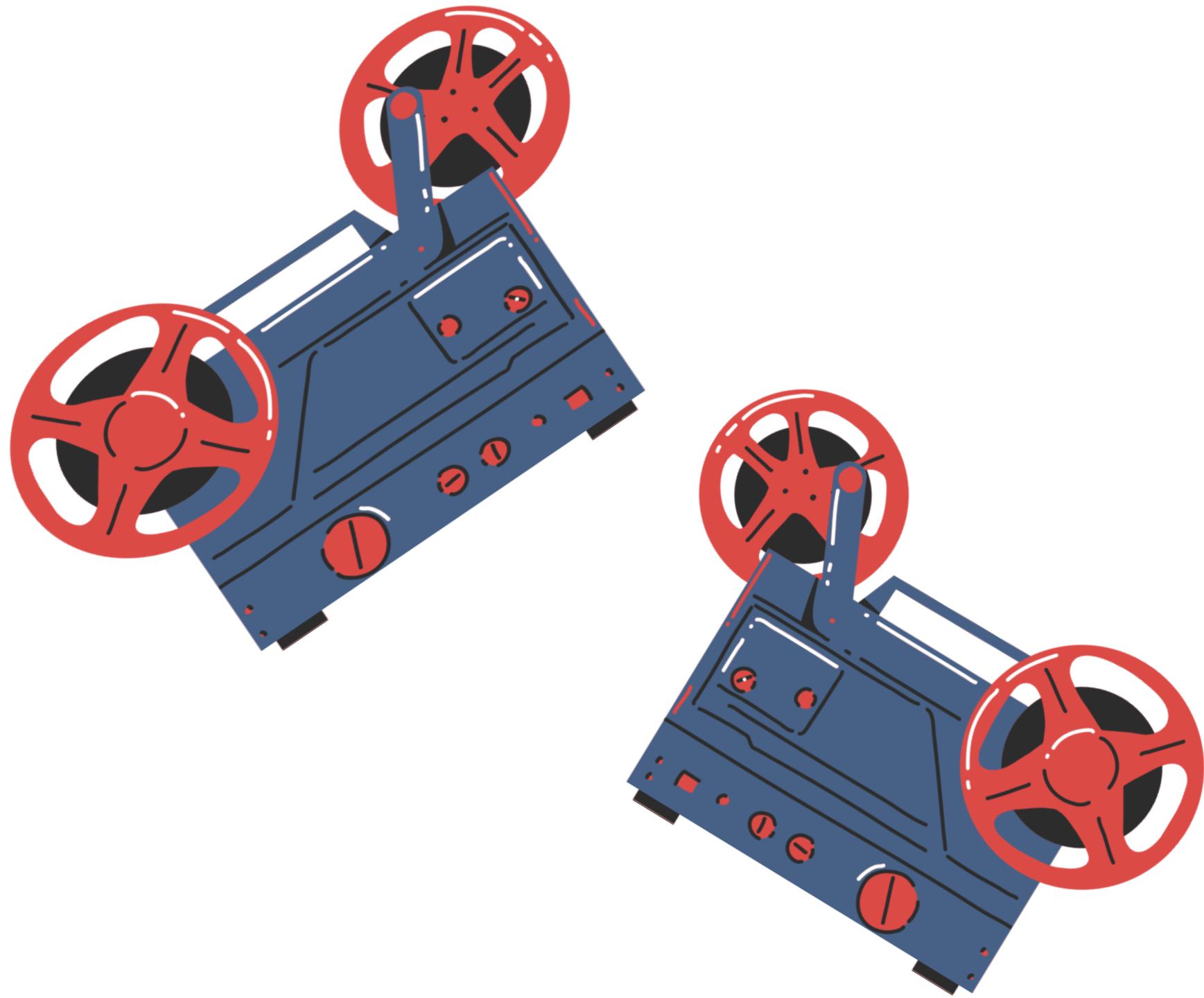
SVD: known for handling large, sparse datasets effectively, SVD uncovers hidden patterns (latent factors) in the way users interact with items, which improves its ability to make recommendations.

KNNBasic: This algorithm uses a straightforward similarity-based approach, recommending items based on the preferences of users with similar tastes.

KNNWithMeans: A more refined version of KNNBasic, this model adjusts predictions by considering the average ratings of users, which often leads to more accurate results.



6. MODELLING EVALUATION



Model Comparison:

SVD: RMSE= 0.8744

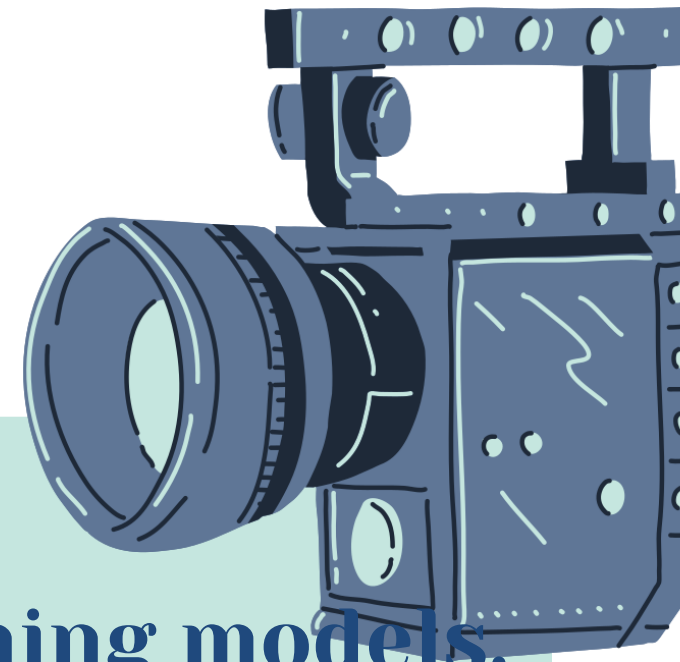
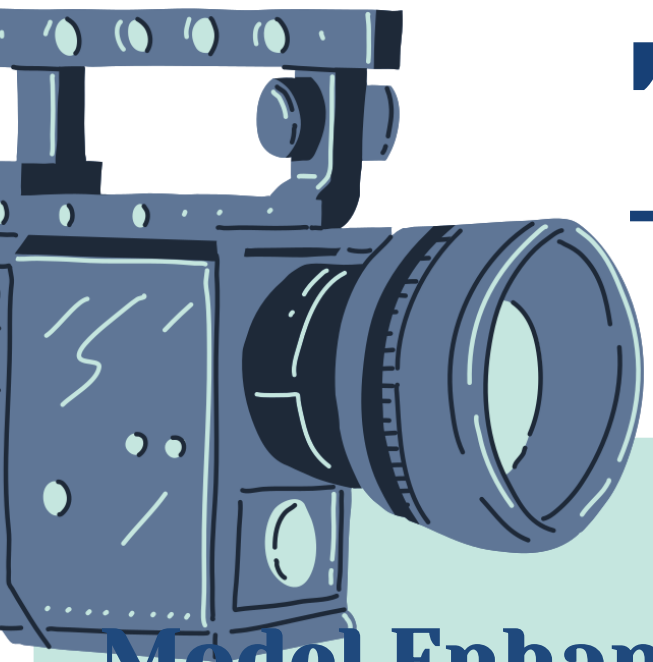
KNN with means: RMSE = 0.9731

KNN Basic: RMSE= 0.8980

Findings:

The SVD Model achieved the lowest RMSE OF 0.8744 indicating that it the best performing model in terms of generating recommendations

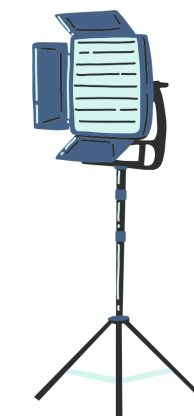
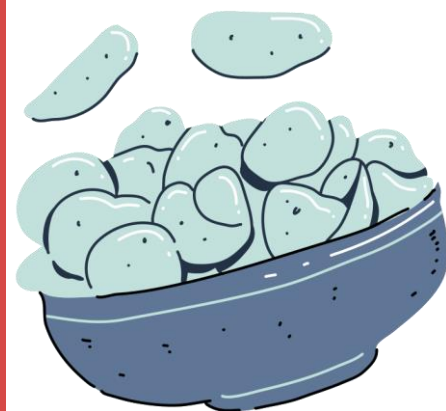
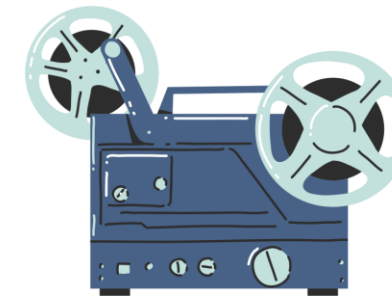
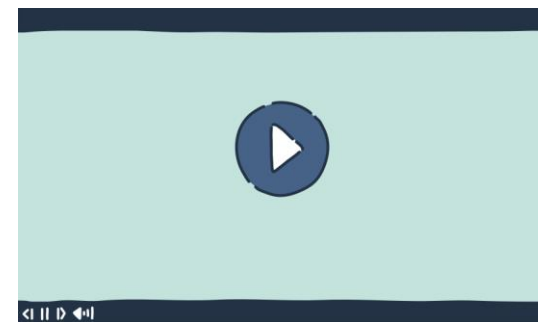
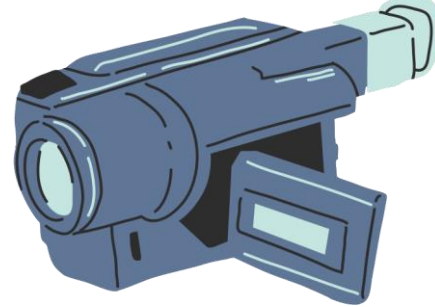
7. RECOMMENDATIONS



Model Enhancements: consider adopting more advanced machine learning models, such as deep learning-based recommendation systems or enhanced matrix factorization techniques. These upgrades can help the system deliver more personalized suggestions, leading to higher user satisfaction.

Data Set Expansion: Broaden the data scope by incorporating a more diverse range of movies, richer user demographic data, and additional user ratings

User Interface Enhancements: Improve the user experience by adding features like movie previews, user reviews, and genre-based filters. These enhancements will create a more interactive and engaging browsing experience,



THANK YOU!!

